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Screen Time, Sleep and Cognition:
A buffering role for sleep in toddlers' cognitive development

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October 22nd, 2026

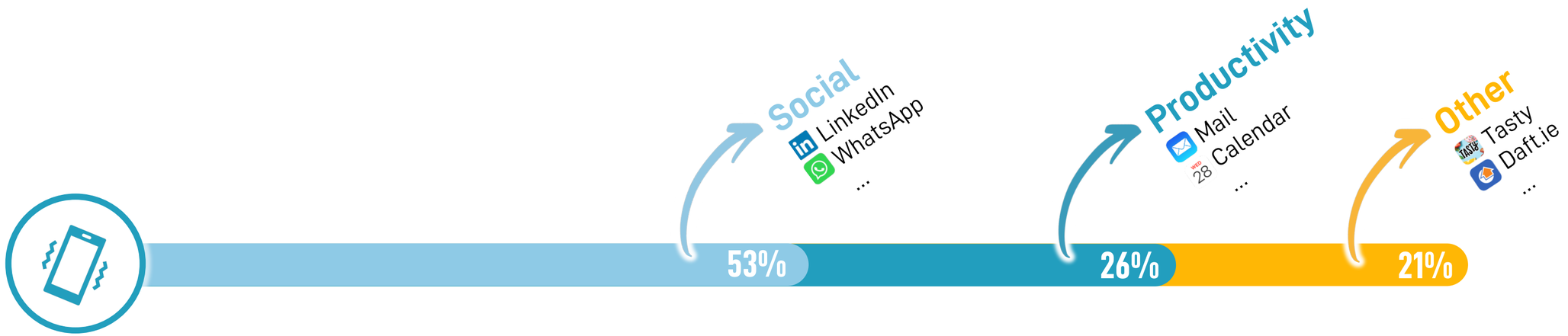
Disclosure of Conflicts of Interest

- I have no conflicts of interest to declare.

Screen Time, Sleep and Cognition

A buffering role for sleep in toddlers' cognitive development

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BACKGROUND

- **Estimates of infant and toddlers' screen time** have surged beyond public health recommendations of *less than 60 min per day*.
- **Executive functions are cognitive skills** that allow control of thoughts, behaviour, and emotions.
- **Given the highly-stimulating nature of digital media**, there are concerns about the impact on the development of executive functions.

The aim of this study was to explore the role of sleep within the association of toddlers' screen time and executive functions.

METHODS

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Data for this study was collected via an **online survey** from February 21st to March 25th, 2025.



Parents & Caregivers

- *N* = 157, 99% Mothers
- 93% Third-Level Education

Island of Ireland

- 26 out of 32 counties
- 29% Dublin (*n* = 46)

Toddlers

- 55% female, 95% White
- 18-36 months
- *M* = 25.4 months (*SD* = 5.47)

METHODS

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Early Executive Functioning Questionnaire

Parent-reported measure of executive function skills in toddlers.

- **Cognitive Executive Functions**
- **Executive Function Regulation**



SCREENS - Questionnaire

Parent-reported measure of toddlers' average, daily screen time.

- **Screen Time**



Brief Infant Sleep Questionnaire

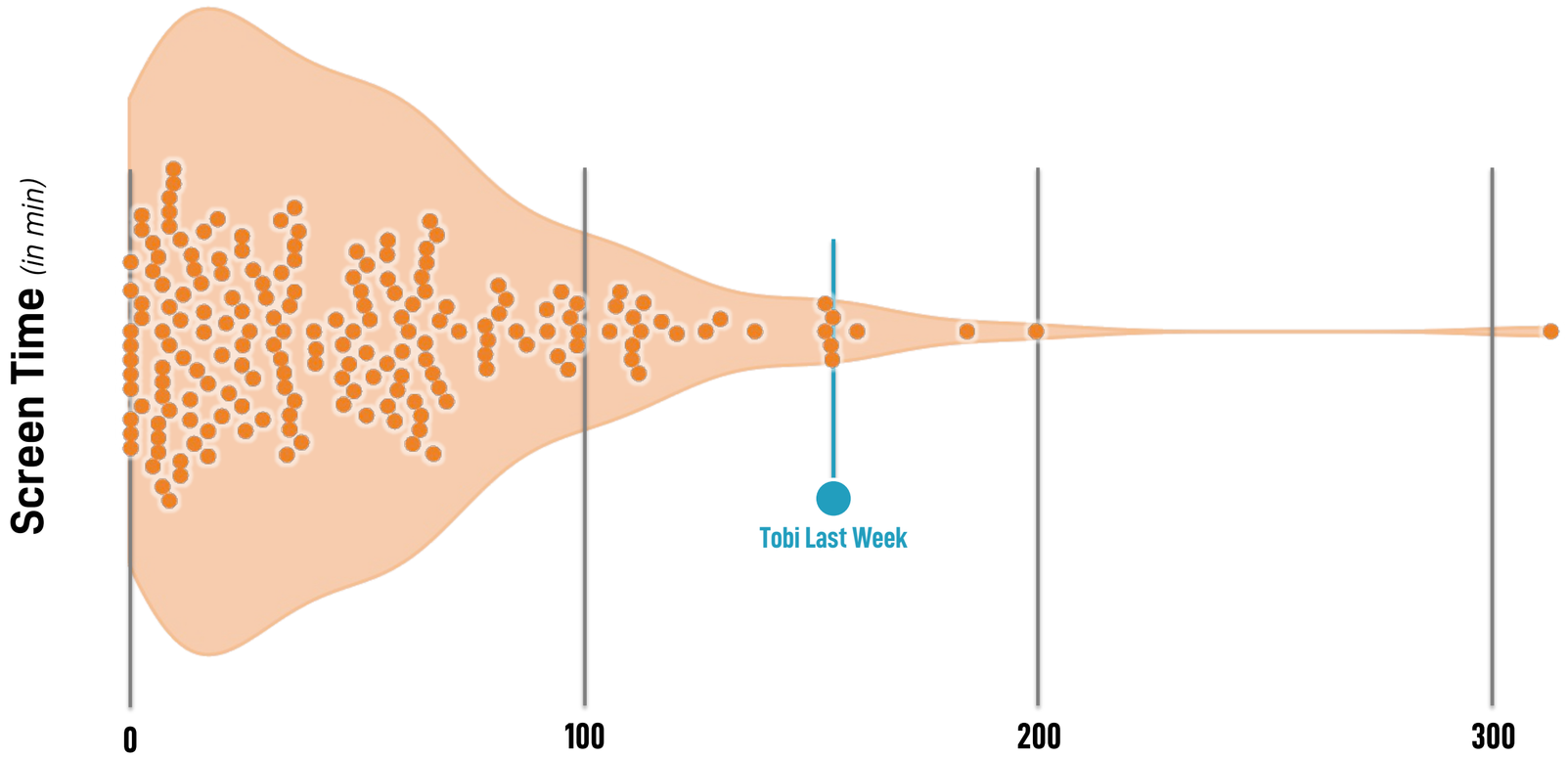
Parent-reported measure of sleep quality in toddlers.

- **Sleep Quality**

RESULTS

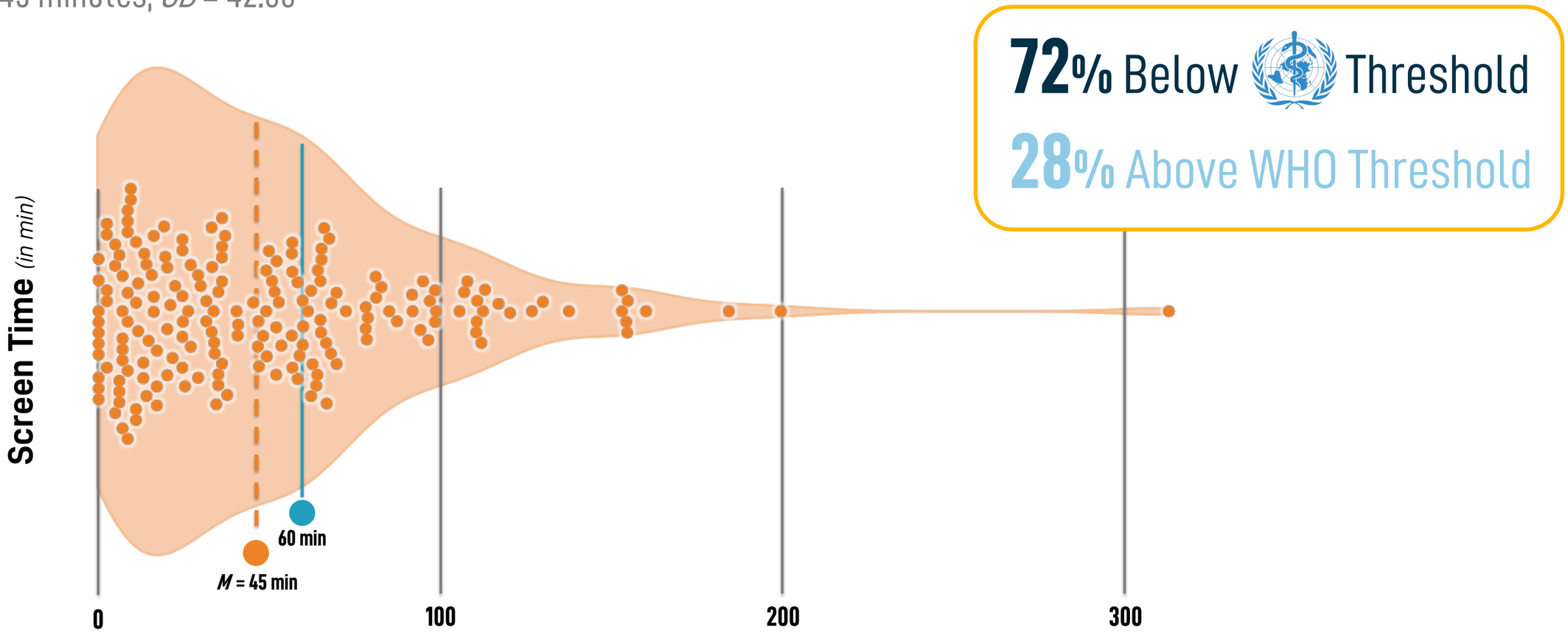
RESULTS

- **What's toddlers' daily screen-time?**



RESULTS

- **What's toddlers' daily screen-time?**
  M = 45.45 minutes, *SD* = 42.88



RESULTS

- **Does your child ask for a specific show or movie regularly?**

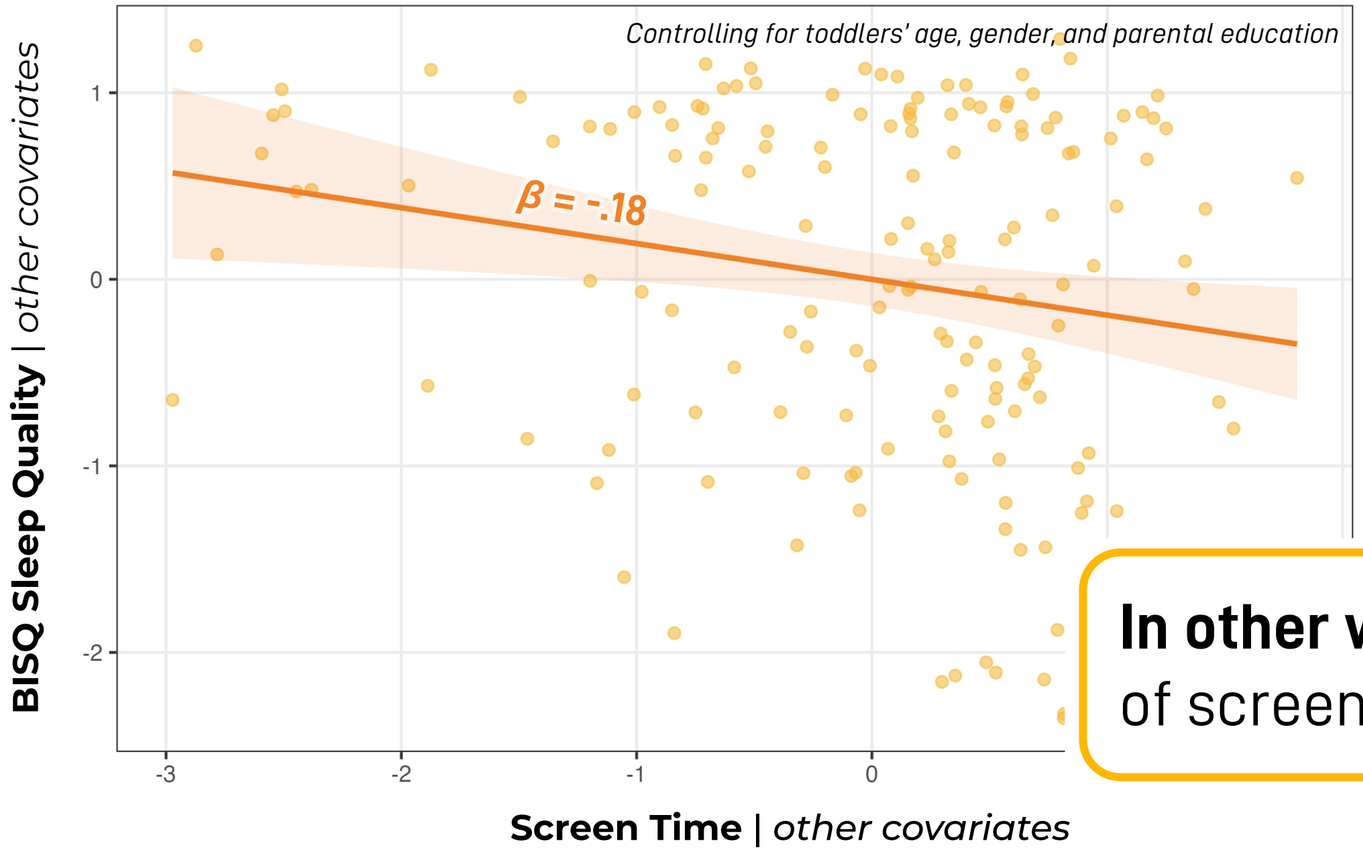
Three out of four toddlers (75%) have a favourite show they are regularly requesting to see.



RESULTS

- **How are screen time and toddlers' sleep quality associated?**

  N = 157

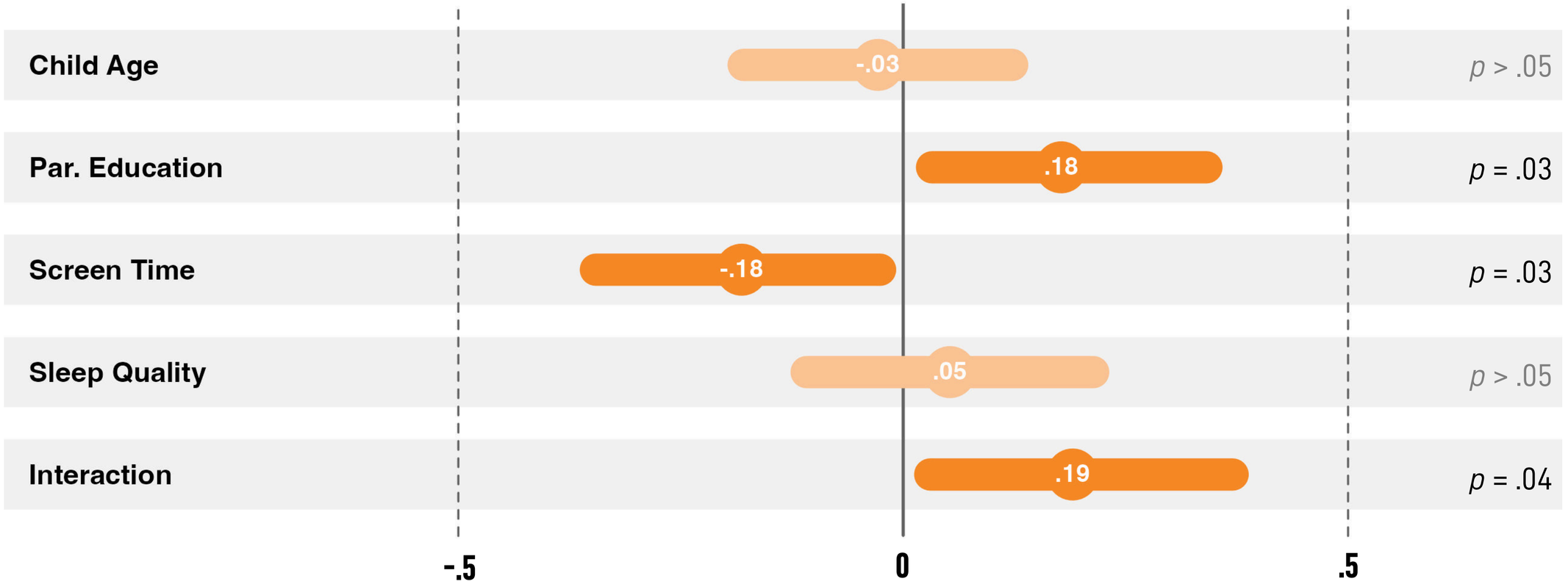


In other words, toddlers with higher amounts of screen time have a lower quality of sleep.

RESULTS

- **Predictors of Toddlers‘ Executive Functions Regulation**

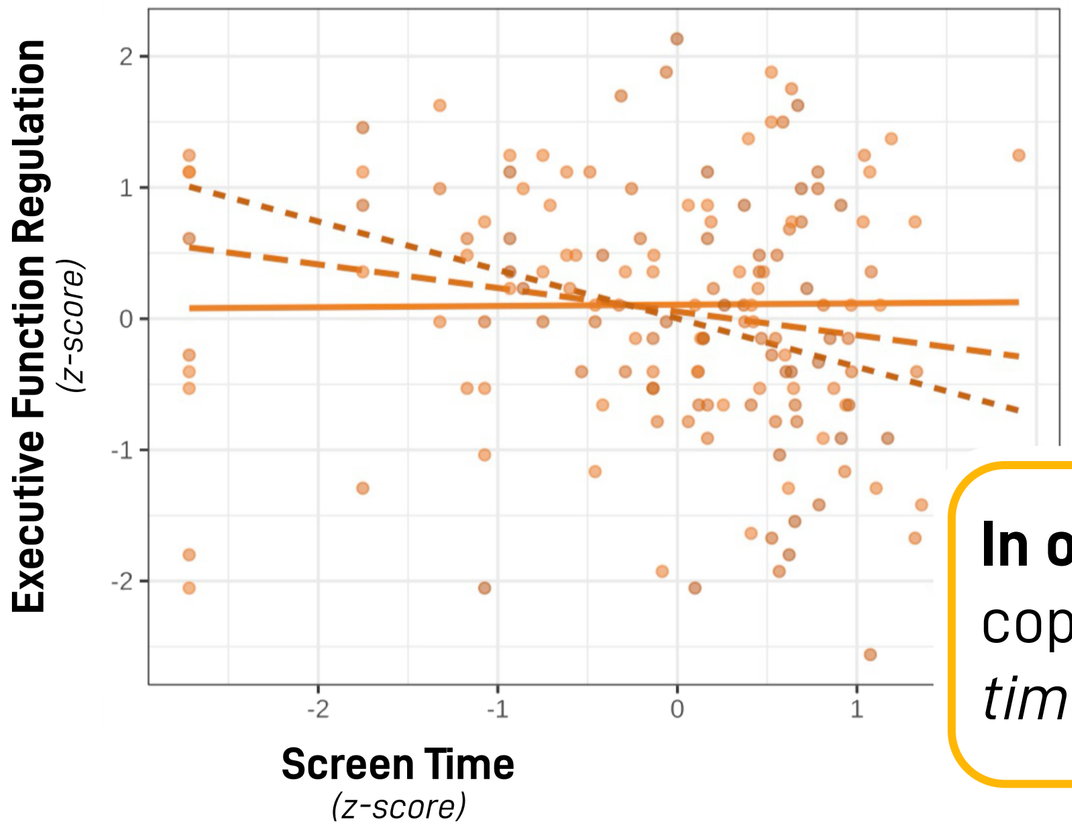
  Results from a multiple regression analysis, $F(5, 151) = 2.63$, $p = .03$, Adj. $R^2 = .05$



Standardised coefficients and their 95% confidence interval are shown.

RESULTS

- **Predictors of Toddlers‘ Executive Functions Regulation**
  Results from a multiple regression analysis, $F(5, 151) = 2.63$, $p = .03$, Adj. $R^2 = .05$

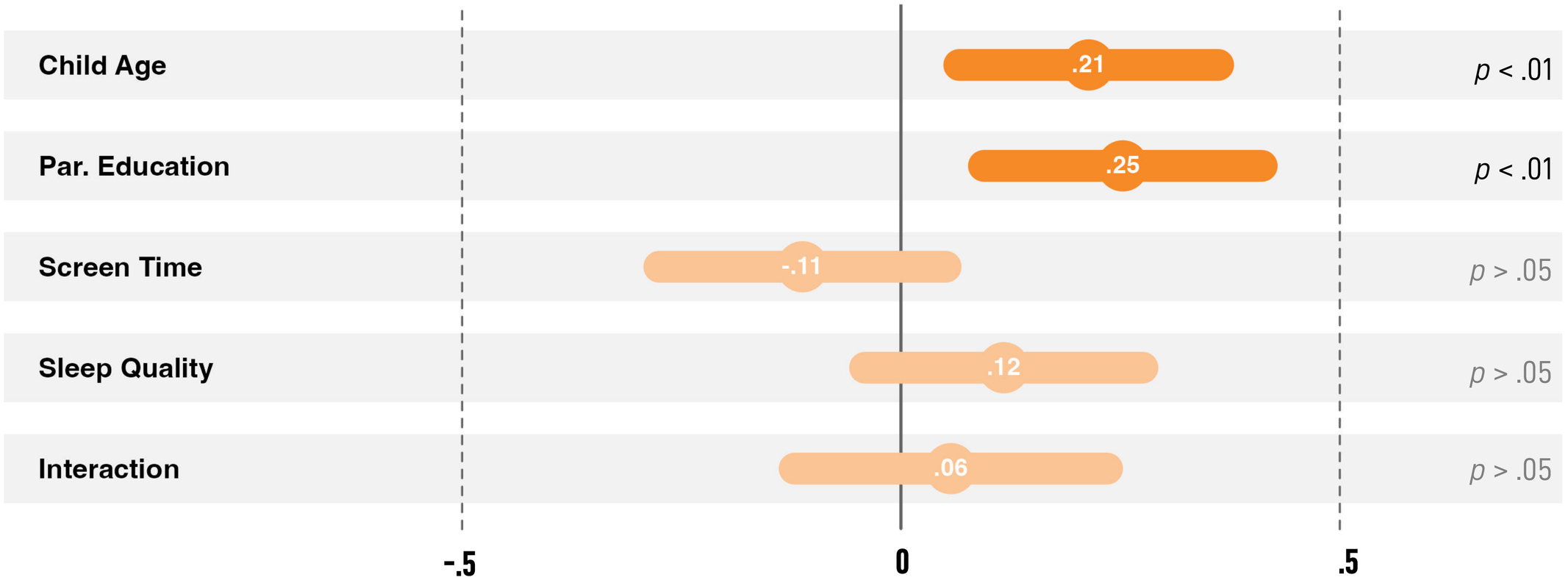


In other words, toddlers who sleep better can cope well with *even long amounts of screen time* compared to their less well-rested peers.

RESULTS

- **Predictors of Toddlers‘ Cognitive Executive Functions**

  Results from a multiple regression analysis, $F(5, 148) = 4.11$, $p = .002$, Adj. $R^2 = .09$



Standardised coefficients and their 95% confidence interval are shown.

TAKE HOME MESSAGE

TAKE HOME MESSAGE

A first estimates of Irish toddlers' screen time.

- Below WHO threshold and similar global estimates.
- Range of attractive digital programming for toddlers.

Sleep buffers the negative role of toddlers' screen time.

- Sleep is negatively associated with screen time.
- Toddlers who are well rested can cope with even long amounts of screen time.

Contextual variables matter when exploring the impact of screen time

STAY IN TOUCH!



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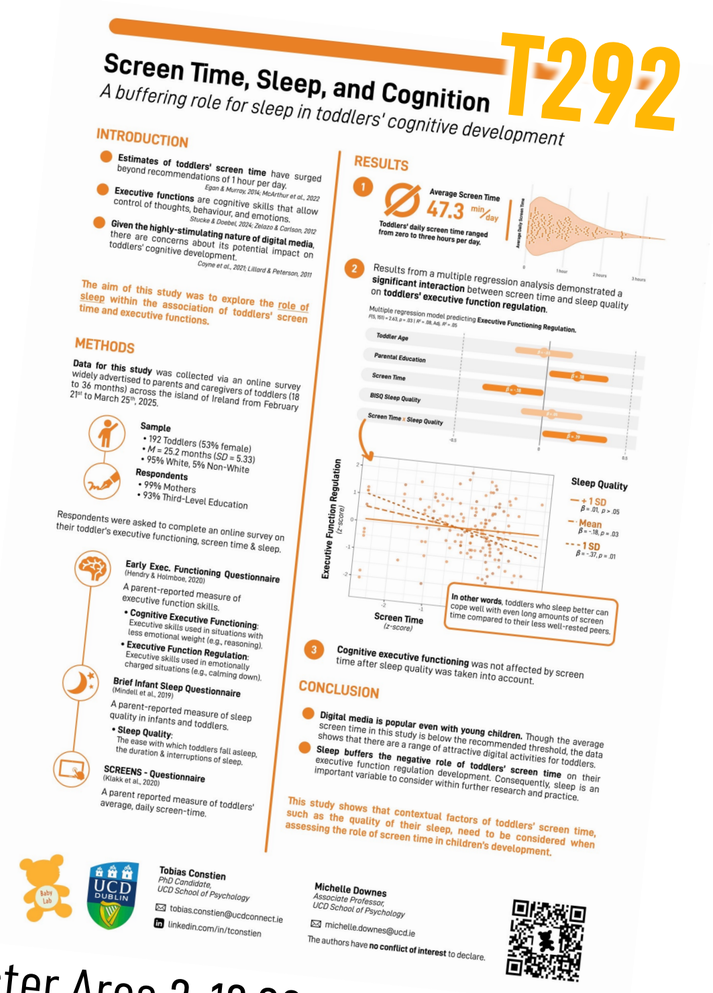
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Poster Area 2, 12:30 – 13:30

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